

COURSE CODE

SBT-DF202

COURSE TITLE

Computer and Digital Forensics

LAB TITLE

USB Image Acquisition and Hash Verification

NAME

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STUDENT ID

2025/FWSD/11325

DATE

30/08/2026

PLATFORM USED

Kali Linux

INSTRUCTOR

Mr. Aminu Idris, AMCPN
Commandant

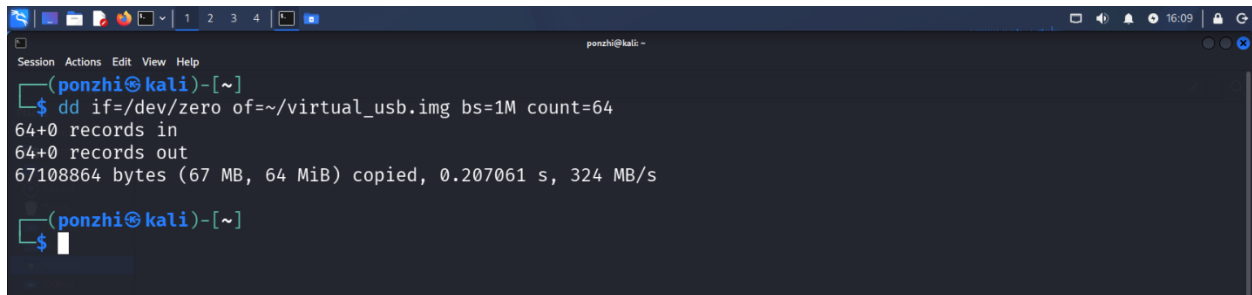
Lab review

By completing this lab, I will demonstrate the ability to explain the purpose of forensic disk imaging, prepare controlled USB evidence, acquire a forensic USB image dd/dcfldd on Linux, calculate MD5 and SHA-256 hash values, compare source and image hashes for integrity verification, validate the contents of the acquired image, identify risks associated with low-level imaging tools, and document the complete acquisition process with professional screenshots and evidence notes.

Linux walkthrough using a loopback virtual disk instead of a physical USB

Create a container file to act as the raw disk

dd if=/dev/zero of=~/.virtual_usb.img bs=1M count=64



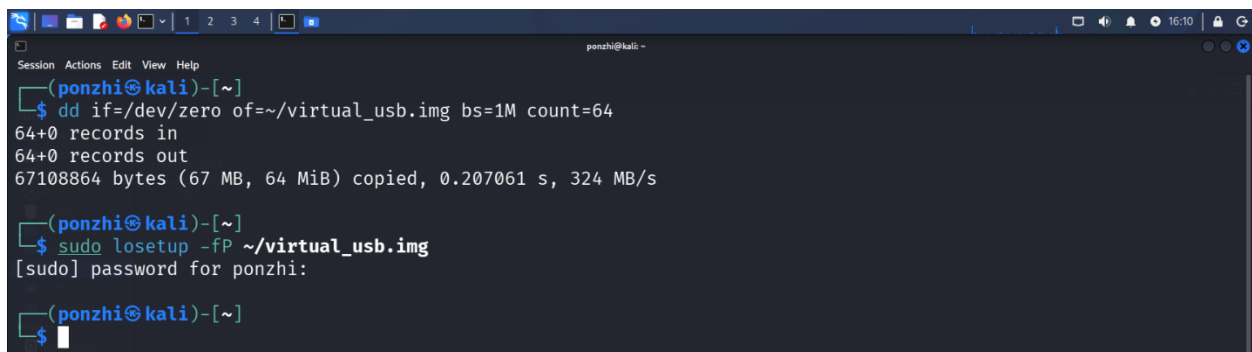
```
ponzhi@kali ~$ dd if=/dev/zero of=~/.virtual_usb.img bs=1M count=64
64+0 records in
64+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 0.207061 s, 324 MB/s

ponzhi@kali ~$
```

- if=/dev/zero: This input provides an endless stream of zero bytes to fill the space.
- of=~/.virtual_usb.img: This is the output file, which will be your virtual disk.
- bs=1M count=64: This command writes in 1MB blocks, a total of 64 times, which makes a 64MB file.

Attach the file as a loop device

sudo losetup -fP ~/.virtual_usb.img



```
ponzhi@kali ~$ dd if=/dev/zero of=~/.virtual_usb.img bs=1M count=64
64+0 records in
64+0 records out
67108864 bytes (67 MB, 64 MiB) copied, 0.207061 s, 324 MB/s

ponzhi@kali ~$ sudo losetup -fP ~/.virtual_usb.img
[sudo] password for ponzhi:

ponzhi@kali ~$
```

- f: automatically finds the first free loop device.
- P: tells the kernel to scan for partitions on it once one exists.

Confirm which loop device you got

losetup -a

```
(ponzhi@kali)-[~]
$ losetup -a
/dev/loop0: []: (/home/ponzhi/virtual_usb.img)

(ponzhi@kali)-[~]
$
```

Partition it

sudo fdisk /dev/loop0

```
(ponzhi@kali)-[~]
$ sudo fdisk /dev/loop0

Welcome to fdisk (util-linux 2.42.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xd01aa763.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p):
```

```
Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xd01aa763.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-131071, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-131071, default 131071):

Created a new partition 1 of type 'Linux' and of size 63 MiB.

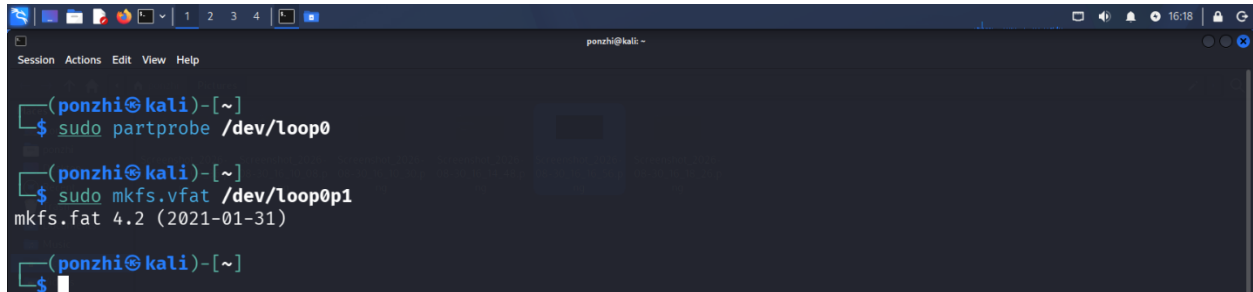
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

(ponzhi@kali)-[~]
$
```

Tell the kernel to re-read the new partition table and format the partition with a filesystem

```
sudo partprobe /dev/loop0
```

```
sudo mkfs.vfat /dev/loop0p1
```

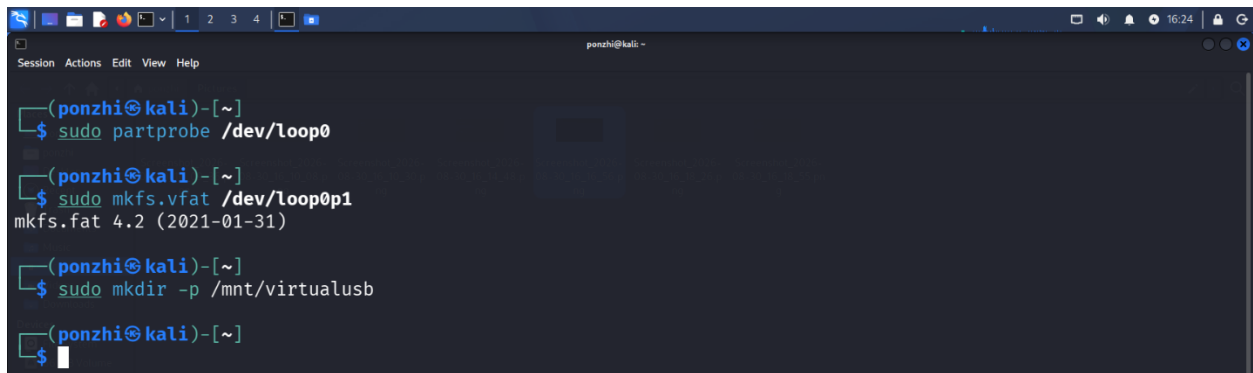
A terminal window titled 'ponzhi@kali' showing the execution of two commands. The first command is 'sudo partprobe /dev/loop0' and the second is 'sudo mkfs.vfat /dev/loop0p1'. The output of the second command is 'mkfs.fat 4.2 (2021-01-31)'.

```
(ponzhi@kali)-[~]  
$ sudo partprobe /dev/loop0  
  
(ponzhi@kali)-[~]  
$ sudo mkfs.vfat /dev/loop0p1  
mkfs.fat 4.2 (2021-01-31)  
  
(ponzhi@kali)-[~]  
$
```

This makes /dev/loop0p1 appear without needing a reboot

Format the partition with a filesystem

```
sudo mkfs.vfat /dev/loop0p1
```

A terminal window titled 'ponzhi@kali' showing the execution of three commands. The first is 'sudo partprobe /dev/loop0', the second is 'sudo mkfs.vfat /dev/loop0p1', and the third is 'sudo mkdir -p /mnt/virtualusb'. The output of the second command is 'mkfs.fat 4.2 (2021-01-31)'.

```
(ponzhi@kali)-[~]  
$ sudo partprobe /dev/loop0  
  
(ponzhi@kali)-[~]  
$ sudo mkfs.vfat /dev/loop0p1  
mkfs.fat 4.2 (2021-01-31)  
  
(ponzhi@kali)-[~]  
$ sudo mkdir -p /mnt/virtualusb  
  
(ponzhi@kali)-[~]  
$
```

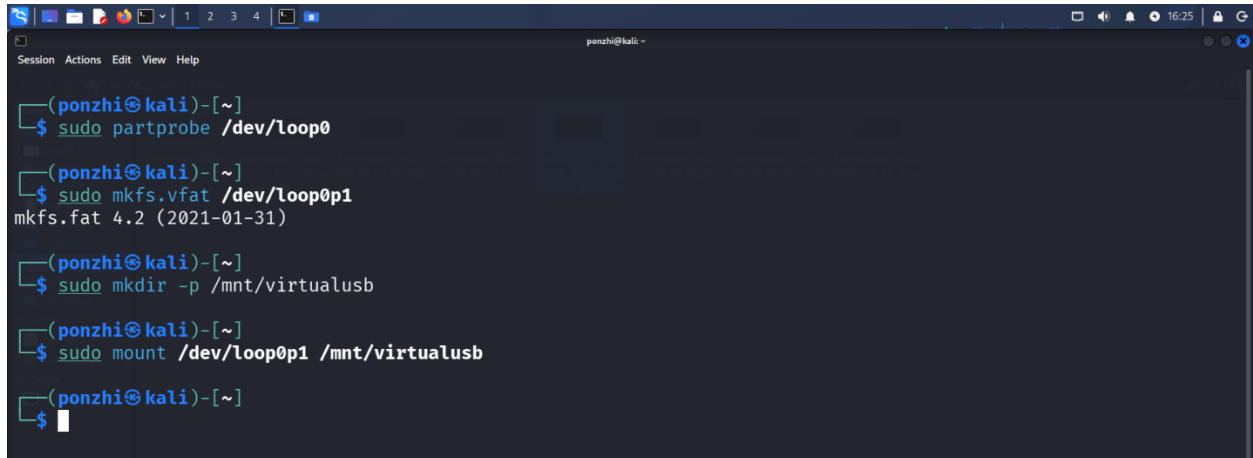
FAT32 mirrors what a typical real USB uses, so my forensic steps closely match the real lab

Part A – Prepare the USB Evidence

Mount the Virtual USB

```
sudo mkdir -p /mnt/virtualusb
```

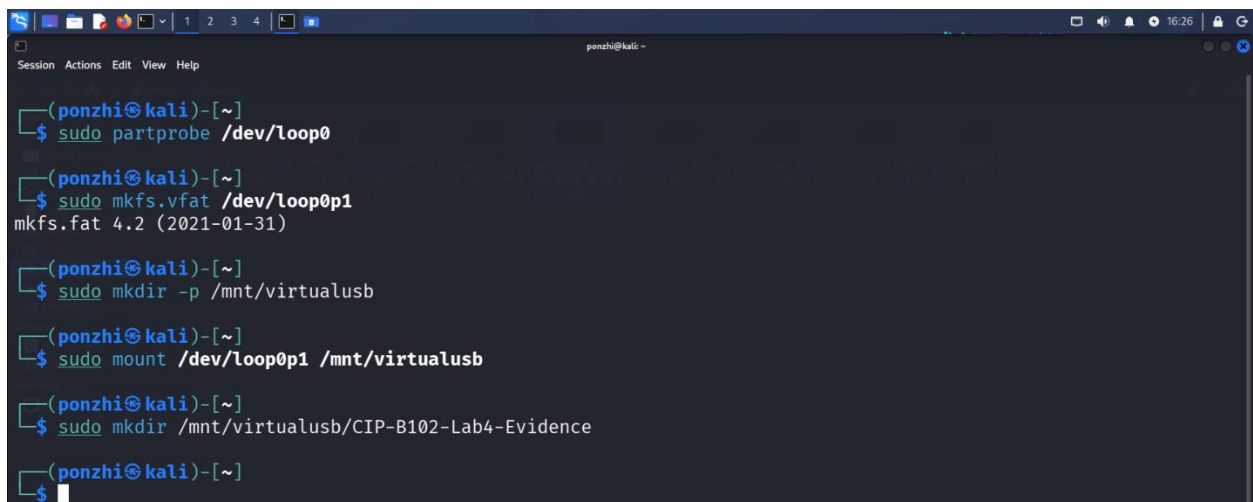
```
sudo mount /dev/loop0p1 /mnt/virtualusb
```

A terminal window titled 'ponzhi@kali' showing a series of commands and their outputs. The commands are: 'sudo partprobe /dev/loop0', 'sudo mkfs.vfat /dev/loop0p1' (output: 'mkfs.fat 4.2 (2021-01-31)'), 'sudo mkdir -p /mnt/virtualusb', and 'sudo mount /dev/loop0p1 /mnt/virtualusb'. The prompt is '(ponzhi@kali)-[~]' and the cursor is at the end of the last command.

```
(ponzhi@kali)-[~]  
$ sudo partprobe /dev/loop0  
  
(ponzhi@kali)-[~]  
$ sudo mkfs.vfat /dev/loop0p1  
mkfs.fat 4.2 (2021-01-31)  
  
(ponzhi@kali)-[~]  
$ sudo mkdir -p /mnt/virtualusb  
  
(ponzhi@kali)-[~]  
$ sudo mount /dev/loop0p1 /mnt/virtualusb  
  
(ponzhi@kali)-[~]  
$
```

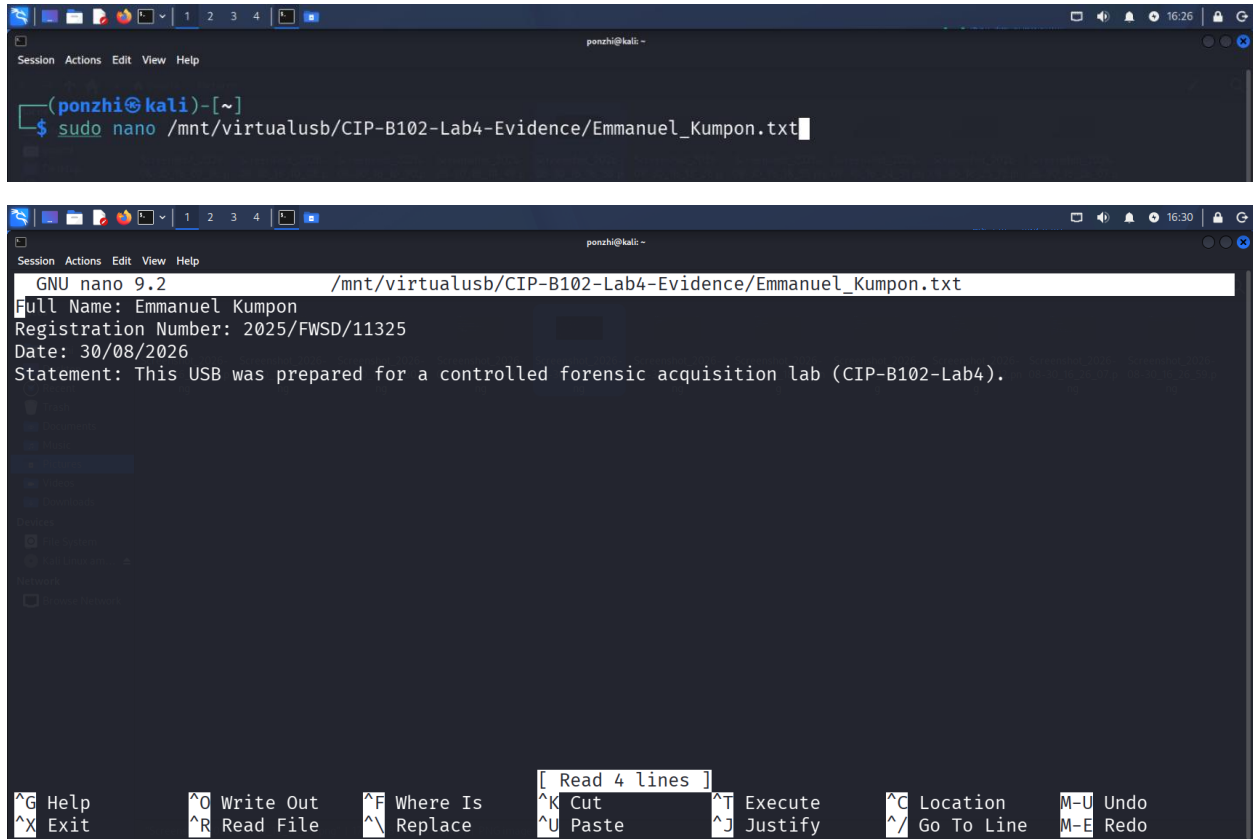
Create the evidence folder and file

```
sudo mkdir /mnt/virtualusb/CIP-B102-Lab4-Evidence
```

A terminal window titled 'ponzhi@kali' showing the same sequence of commands as the previous screenshot, followed by 'sudo mkdir /mnt/virtualusb/CIP-B102-Lab4-Evidence'. The prompt is '(ponzhi@kali)-[~]' and the cursor is at the end of the last command.

```
(ponzhi@kali)-[~]  
$ sudo partprobe /dev/loop0  
  
(ponzhi@kali)-[~]  
$ sudo mkfs.vfat /dev/loop0p1  
mkfs.fat 4.2 (2021-01-31)  
  
(ponzhi@kali)-[~]  
$ sudo mkdir -p /mnt/virtualusb  
  
(ponzhi@kali)-[~]  
$ sudo mount /dev/loop0p1 /mnt/virtualusb  
  
(ponzhi@kali)-[~]  
$ sudo mkdir /mnt/virtualusb/CIP-B102-Lab4-Evidence  
  
(ponzhi@kali)-[~]  
$
```

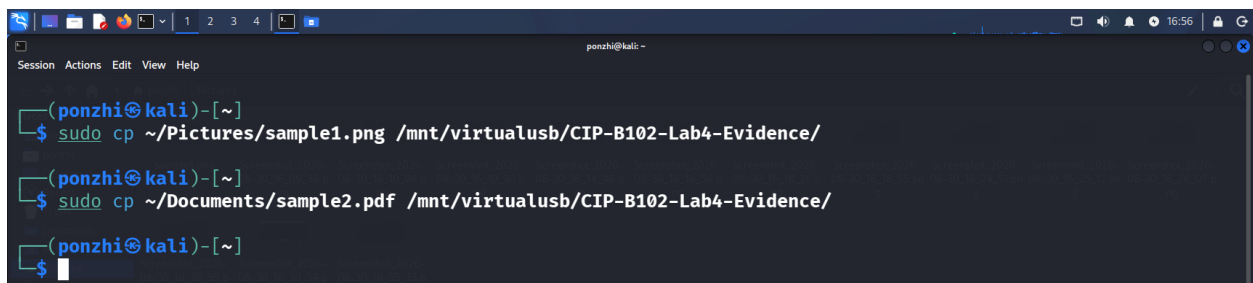
sudo nano /mnt/virtualusb/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt



```
(ponzhi@kali)-[~]
$ sudo nano /mnt/virtualusb/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt

GNU nano 9.2 /mnt/virtualusb/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt
Full Name: Emmanuel Kumpon
Registration Number: 2025/FWSD/11325
Date: 30/08/2026
Statement: This USB was prepared for a controlled forensic acquisition lab (CIP-B102-Lab4).
```

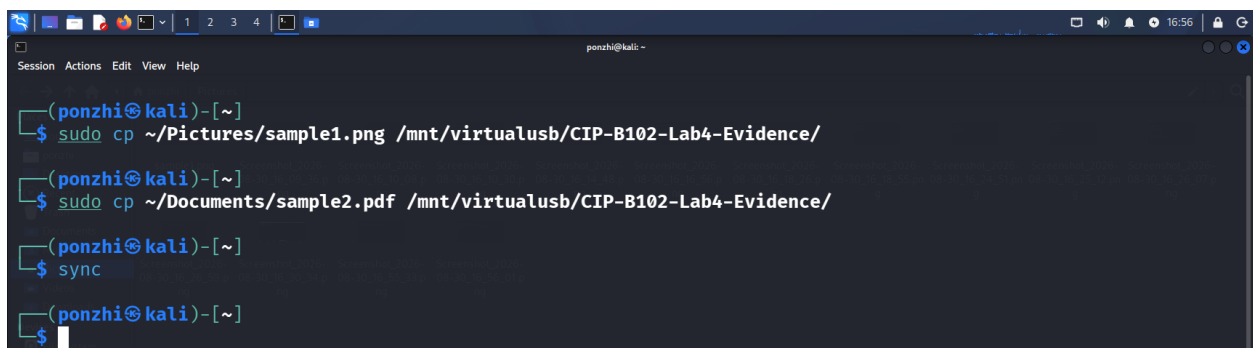
Add one image and one PDF



```
(ponzhi@kali)-[~]
$ sudo cp ~/Pictures/sample1.png /mnt/virtualusb/CIP-B102-Lab4-Evidence/
(ponzhi@kali)-[~]
$ sudo cp ~/Documents/sample2.pdf /mnt/virtualusb/CIP-B102-Lab4-Evidence/
(ponzhi@kali)-[~]
$
```

Flush writes to disk

sync



```
(ponzhi@kali)-[~]
$ sudo cp ~/Pictures/sample1.png /mnt/virtualusb/CIP-B102-Lab4-Evidence/
(ponzhi@kali)-[~]
$ sudo cp ~/Documents/sample2.pdf /mnt/virtualusb/CIP-B102-Lab4-Evidence/
(ponzhi@kali)-[~]
$ sync
(ponzhi@kali)-[~]
$
```

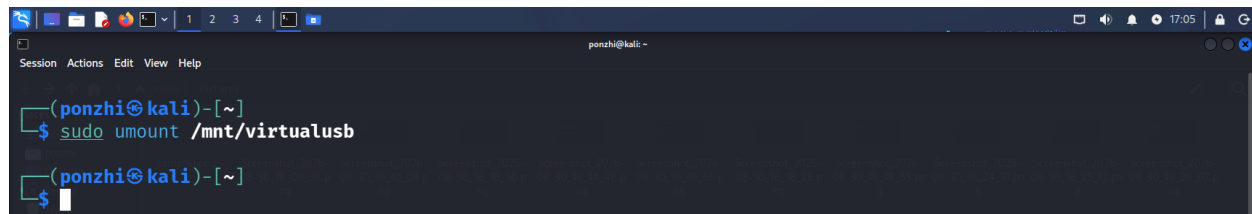
Part B - Complete Evidence Documentation

Case / Lab ID	CIP-B102-Lab4
Student name	Emmanuel Kumpon
USB brand/description	Simulated USB drive (loopback virtual disk, /dev/loop0)
USB capacity	64MB
Acquisition method	dd (Linux), loop device substituted for physical USB
Tool and version	GNU coreutils dd; fdisk
Image filename	CIPB102_Lab4_Emmanuel_Kumpon_USB.dd
Image storage location	~/CIP-B102-Lab4
Hash algorithms used	MD5, SHA-256

Part C - Acquire the USB Image

Unmount the partition first

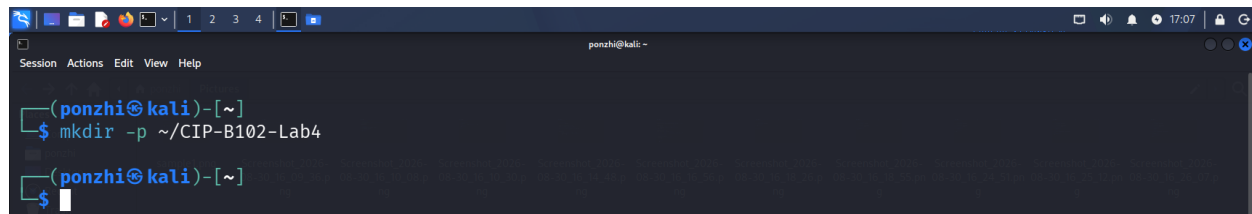
sudo umount /mnt/virtualusb

A terminal window with a dark background and light blue text. The prompt is '(ponzhi@kali)-[~]'. The command 'sudo umount /mnt/virtualusb' has been entered and executed. The prompt returns to '(ponzhi@kali)-[~]'.

```
(ponzhi@kali)-[~]  
$ sudo umount /mnt/virtualusb  
(ponzhi@kali)-[~]  
$
```

Create the destination folder on your real disk

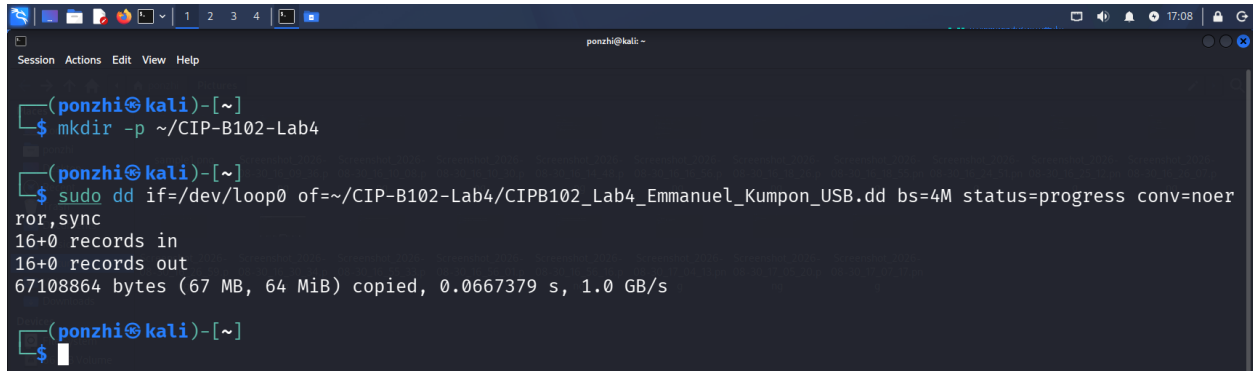
mkdir -p ~/CIP-B102-Lab4

A terminal window with a dark background and light blue text. The prompt is '(ponzhi@kali)-[~]'. The command 'mkdir -p ~/CIP-B102-Lab4' has been entered and executed. The prompt returns to '(ponzhi@kali)-[~]'.

```
(ponzhi@kali)-[~]  
$ mkdir -p ~/CIP-B102-Lab4  
(ponzhi@kali)-[~]  
$
```

Run the acquisition image on the whole loop device, not just the partition

```
sudo dd if=/dev/loop0 of=~/.CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd  
bs=4M status=progress conv=noerror,sync
```

A terminal window titled 'ponzhi@kali' showing the execution of the dd command. The user first creates a directory '~/.CIP-B102-Lab4' with 'mkdir -p'. Then, they run 'sudo dd if=/dev/loop0 of=~/.CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd bs=4M status=progress conv=noerror,sync'. The output shows '16+0 records in', '16+0 records out', and '67108864 bytes (67 MB, 64 MiB) copied, 0.0667379 s, 1.0 GB/s'.

```
(ponzhi@kali)~  
$ mkdir -p ~/.CIP-B102-Lab4  
  
(ponzhi@kali)~  
$ sudo dd if=/dev/loop0 of=~/.CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd bs=4M status=progress conv=noerror,sync  
16+0 records in  
16+0 records out  
67108864 bytes (67 MB, 64 MiB) copied, 0.0667379 s, 1.0 GB/s  
  
(ponzhi@kali)~  
$
```

if=/dev/loop0: The entire virtual device, the loopback equivalent of /dev/sdb.

bs=4M: 4MB block size for speed.

status=progress: live progress display.

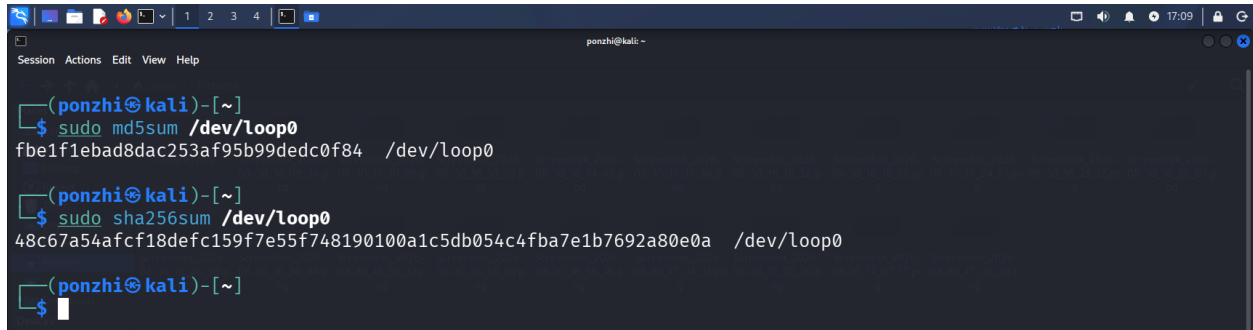
conv=noerror, sync: keep going past unreadable blocks, padding with zeros to preserve alignment.

Part D - Hash Verification

Hash the source

```
sudo md5sum /dev/loop0
```

```
sudo sha256sum /dev/loop0
```

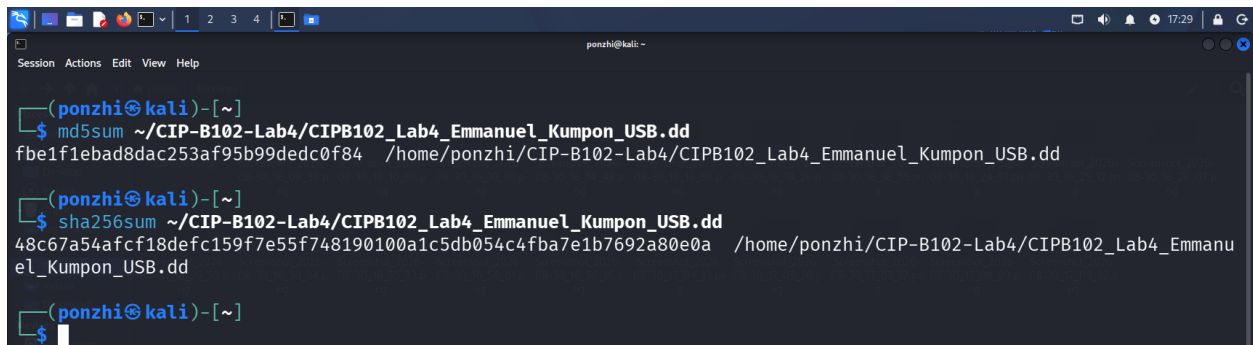
A terminal window titled 'ponzhi@kali' showing the execution of two commands. The first command is 'sudo md5sum /dev/loop0', which outputs 'fbe1f1ebad8dac253af95b99dedc0f84 /dev/loop0'. The second command is 'sudo sha256sum /dev/loop0', which outputs '48c67a54afc18defc159f7e55f748190100a1c5db054c4fba7e1b7692a80e0a /dev/loop0'. The prompt is '(ponzhi@kali)-[~]' and the cursor is on a new line.

```
(ponzhi@kali)-[~]  
$ sudo md5sum /dev/loop0  
fbe1f1ebad8dac253af95b99dedc0f84 /dev/loop0  
  
(ponzhi@kali)-[~]  
$ sudo sha256sum /dev/loop0  
48c67a54afc18defc159f7e55f748190100a1c5db054c4fba7e1b7692a80e0a /dev/loop0  
  
(ponzhi@kali)-[~]  
$
```

Hash the acquired image file

```
md5sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd
```

```
sha256sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd
```

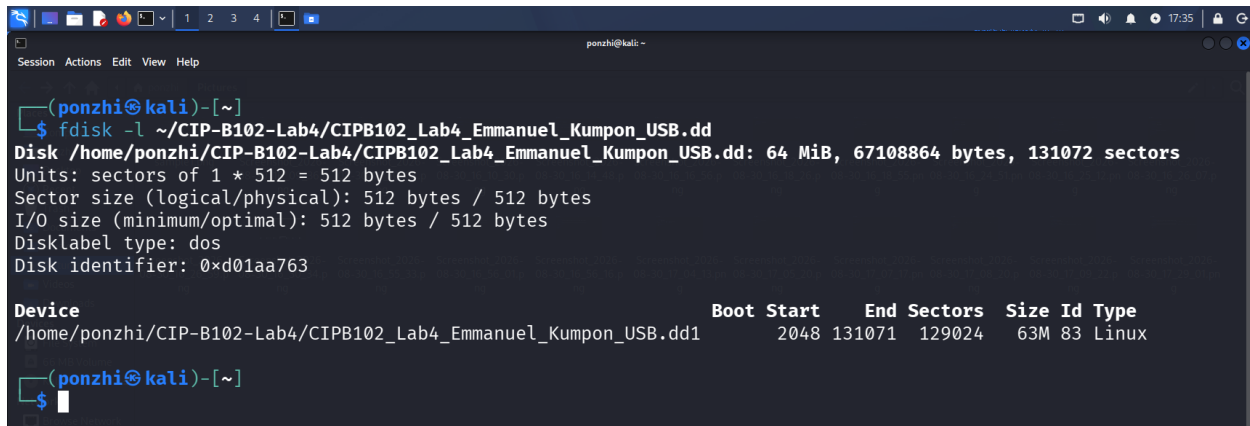
A terminal window titled 'ponzhi@kali' showing the execution of two commands. The first command is 'md5sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd', which outputs 'fbe1f1ebad8dac253af95b99dedc0f84 /home/ponzhi/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd'. The second command is 'sha256sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd', which outputs '48c67a54afc18defc159f7e55f748190100a1c5db054c4fba7e1b7692a80e0a /home/ponzhi/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd'. The prompt is '(ponzhi@kali)-[~]' and the cursor is on a new line.

```
(ponzhi@kali)-[~]  
$ md5sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd  
fbe1f1ebad8dac253af95b99dedc0f84 /home/ponzhi/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd  
  
(ponzhi@kali)-[~]  
$ sha256sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd  
48c67a54afc18defc159f7e55f748190100a1c5db054c4fba7e1b7692a80e0a /home/ponzhi/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd  
  
(ponzhi@kali)-[~]  
$
```

Part E - Validate the Acquired Image

Find the partition's byte offset inside the image

`fdisk -l ~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd`



A terminal window titled 'ponzhi@kali' showing the output of the `fdisk -l` command. The output displays disk information for `/home/ponzhi/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd`, including its size (64 MiB, 67108864 bytes, 131072 sectors), sector size (512 bytes), and I/O size (512 bytes). It also shows the disk label type as 'dos' and the disk identifier as `0xd01aa763`. A table lists the disk's partitions:

Device	Boot	Start	End	Sectors	Size	Id	Type
<code>/home/ponzhi/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd1</code>		2048	131071	129024	63M	83	Linux

Calculate the offset

$\text{offset} = \text{start_sector} \times \text{sector_size}$

$\text{start_sector} = 2048$

$\text{sector_size} = 512$

$\text{Offset} = 2048 \times 512 = 1,048,576$

Mount the image read-only through loopback

`mkdir ~/image_mount`

`sudo mount -o ro,loop,offset=1048576 ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd ~/image_mount`

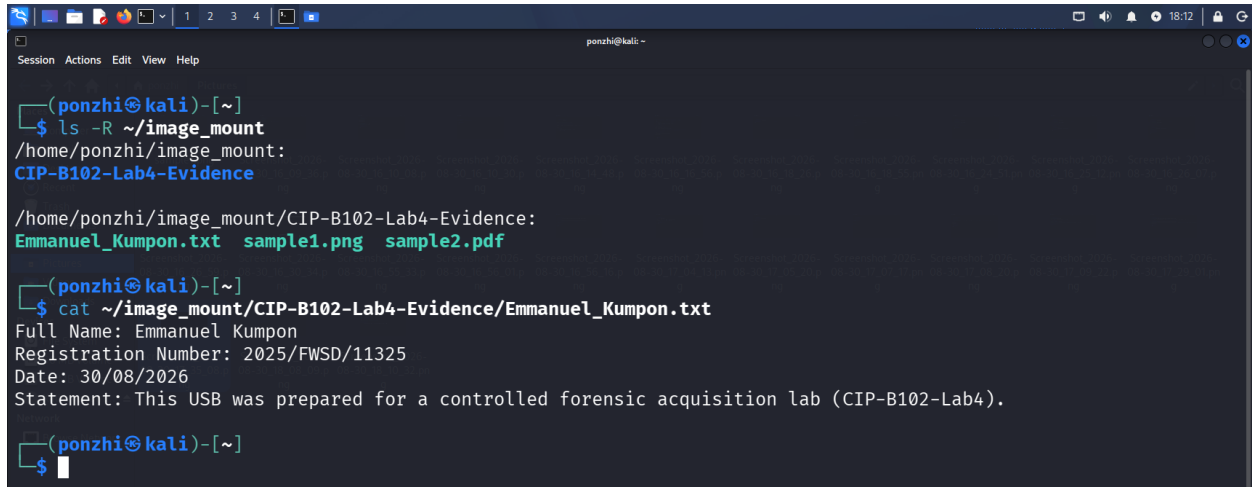


A terminal window titled 'ponzhi@kali' showing the execution of two commands. First, `mkdir ~/image_mount` is run to create the mount directory. Then, `sudo mount -o ro,loop,offset=1048576 ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd ~/image_mount` is run to mount the disk image read-only using a loopback device. The terminal shows the password prompt for 'ponzhi' and a successful prompt for the next command.

Confirm the evidence file is present and intact

```
ls -R ~/image_mount
```

```
cat ~/image_mount/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt
```

A screenshot of a terminal window titled 'ponzhi@kali: ~'. The terminal shows the following commands and output:
1. Command: `ls -R ~/image_mount`
Output: `/home/ponzhi/image_mount:`
`CIP-B102-Lab4-Evidence`
`/home/ponzhi/image_mount/CIP-B102-Lab4-Evidence:`
`Emmanuel_Kumpon.txt sample1.png sample2.pdf`
2. Command: `cat ~/image_mount/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt`
Output: `Full Name: Emmanuel Kumpon`
`Registration Number: 2025/FWSD/11325`
`Date: 30/08/2026`
`Statement: This USB was prepared for a controlled forensic acquisition lab (CIP-B102-Lab4).`
The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The top status bar shows 'ponzhi@kali: ~' and system icons on the right.

Conclusion

We completed the acquisition process using a loopback virtual disk instead of a physical USB device. The process included preparing evidence, documenting steps before the acquisition, creating a raw image with `dd`, checking the cryptographic hash, and verifying the image content in read-only mode. This process follows the standard procedures for acquiring data from a physical USB and shows that we have the necessary skills for this lab.